Kinematics Review

Constant Acceleration

With a constant acceleration the velocity changes by the same amount in every second. This is the *simplest* kind of changing motion, and it is a good model for a cart rolling down a smooth ramp.

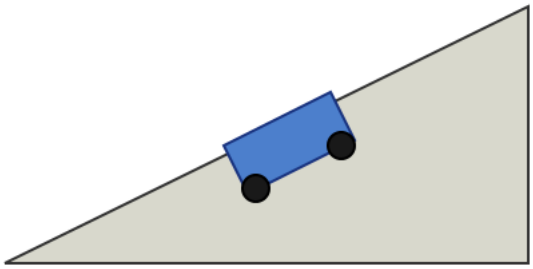
Worked example

A cart starts from rest and accelerates at two metres per second squared for three seconds. Its final velocity follows from the definition of acceleration:

$$v_f = v_0 + at$$
$$= 0 + 2 \times 3 = 6$$

The answer has units of metres per second, which is a useful check on the working.

Figure 3 A cart on a ramp, drawn as a simple block on a slope.



The ramp in the figure is smooth, so friction can be ignored. Here $a = 2$ m/s^2 and $t = 3$ s, typed as ordinary text rather than as an equation.

Summary

Constant acceleration links velocity, time and distance with a few short equations.